

vla-eval — A Unified Evaluation Harness for Vision-Language-Action Models

Suhwan Choi^{1*} Yunsung Lee¹ Yubeen Park¹ Chris Dongjoo Kim² Ranjay Krishna² Dieter Fox² Youngjae Yu³

¹MAUM.AI ²Allen Institute for AI (AI2) ³Seoul National University *primary author



47×

FASTER EVALUATION

O(N+M)

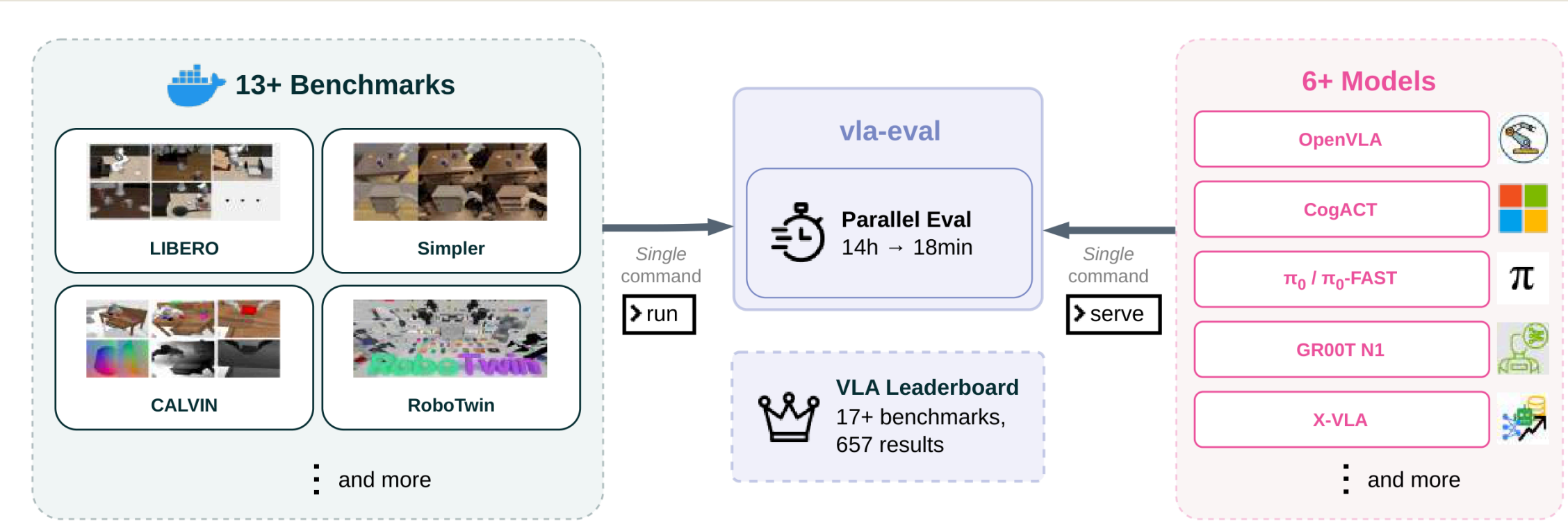
INTEGRATION COST

14×6

BENCHMARK × MODEL MATRIX

657

LEADERBOARD RESULTS



Integrate **once**.
Evaluate **everywhere**.

14 benchmarks & 6 model servers integrate a single time — the full cross-evaluation matrix then runs from two commands.

01 THE PROBLEM

Per-benchmark integration doesn't scale

VLA models are evaluated on **many simulation benchmarks** — but wiring each one in is heavy, repeated engineering.

- ◆ **Duplicated effort.** Every model repo keeps its own eval scripts, drifting apart silently.
- ◆ **Dependency conflicts.** LIBERO needs Python 3.8, ManiSkill2 needs 3.10, CALVIN needs PyBullet — no shared environment.
- ◆ **Underspecified protocols.** Seeds, episode counts and preprocessing are omitted from papers.

O(N×M)

Naïve cost — re-integrate for **every** model × benchmark pair.



O(N+M)

Integrate each **once** — the full N×M matrix then **fills itself**.

vla-eval

02 FRAMEWORK — SETUP

Two commands, zero dependency hell

- ◆ **Dependency isolation.** PEP 723 metadata + `uv run`; conflicting versions coexist.
- ◆ **Declarative.** Two YAML configs; Docker images bundle every asset.
- ◆ **Reproducible.** Each run emits structured JSON — version, config, metrics.

```
$ vla-eval serve --config model.yaml
$ vla-eval run --config benchmark.yaml
```

14 BENCHMARKS

SimplerEnv LIBERO CALVIN RLBench LIBERO-Pro
RoboCerebra ManiSkill2 Kinetix MIKASA LIBERO-Mem
RoboMME VLABench RoboTwin 2.0 RoboCasa

6 MODEL SERVERS

OpenVLA OpenVLA-OFT CogACT π_0 / FAST GR00T N1 X-VLA

04 VALIDATION

Reproduce six published VLA codebases

Across **6 VLA codebases** and **3 benchmarks**, with fixed seeds and versioned Docker images.

CODEBASE	LIBERO %	CALVIN	SIMPLER %
OpenVLA	76.2 ± 0.3	—	—
$\pi_{0.5}$	97.7 ± 0.8	—	—
OpenVLA-OFT	96.7 ± 0.4	—	—
GR00T N1.6	94.9 ± 2.1	—	59.7 ± 8.0
DB-CogACT	94.7 ± 0.2	4.02 ± 0.04	63.5 ± 6.0
X-VLA	97.4 ± 0.7	4.30 ± 1.13	94.8 ± 1.0

Ours, with Δ vs. published. — = no public checkpoint.

02 FRAMEWORK DESIGN

Decouple the model from the benchmark

Inspired by the decoupled design of `lm-evaluation-harness` for LLMs, vla-eval splits model inference from benchmark execution over a **WebSocket + msgpack** protocol — and isolates each benchmark inside its own **Docker** container.

- ◆ **Models integrate once.** One blocking `predict()` — ~50 lines.
- ◆ **Benchmarks integrate once.** Four methods inside a pinned Docker image.

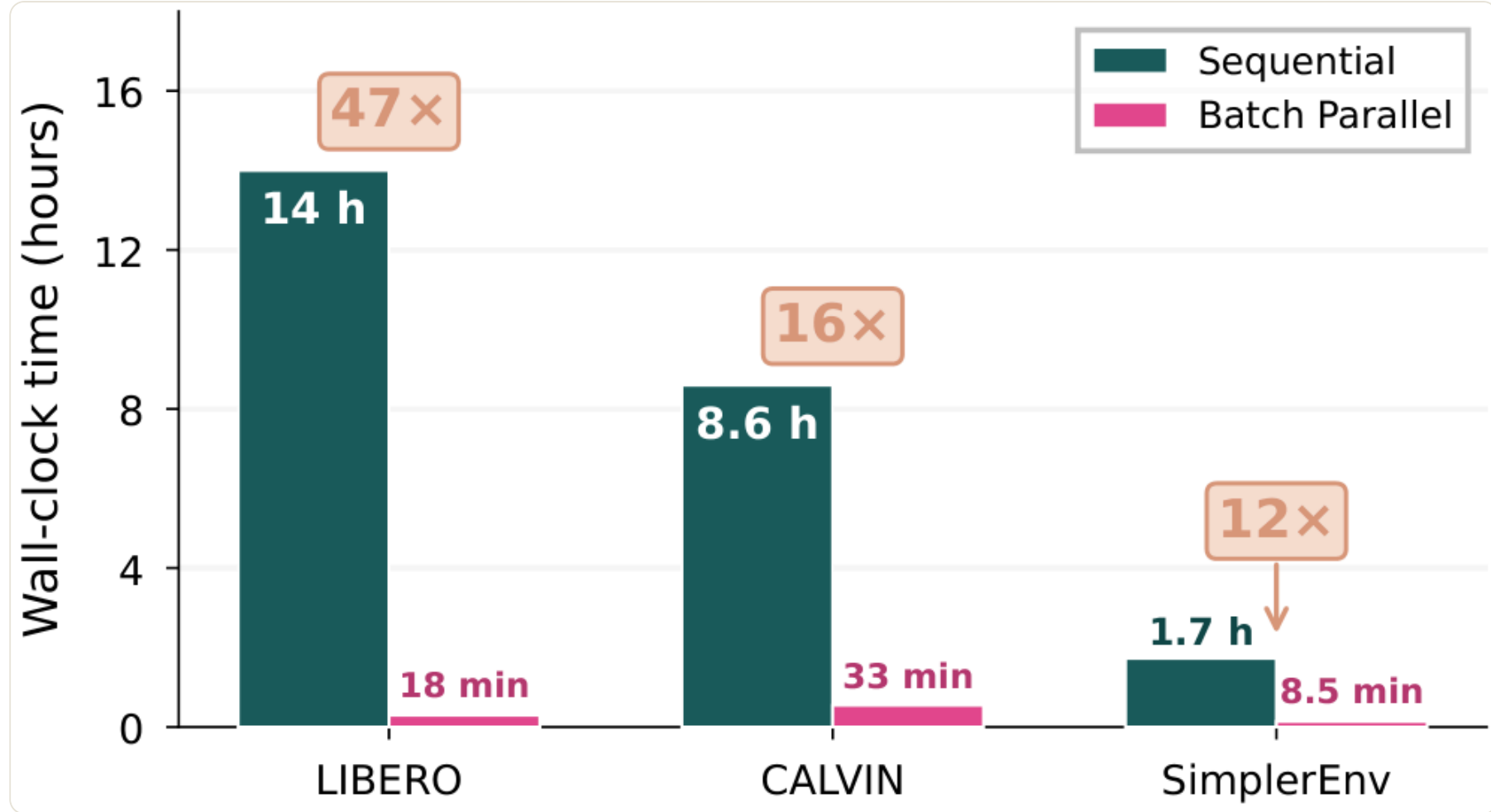
```
class OpenVLAserver(PredictModelServer):  
    def predict(self, obs, ctx):  
        img = Image.fromarray(  
            next(iter(obs["images"].values()))  
        )  
        prompt = f"In: action for {obs['task']}? Out:"  
        inp = self._proc(prompt, img).to("cuda")  
        act = self._model.predict_action(**inp)  
        return {"actions": act}
```

MODEL SIDE, IN FULL

03 PARALLEL EVALUATION

47× faster — and model-agnostic

Environment parallelism via **episode sharding** across K Docker containers; inference parallelism via **batched forward passes**.



Wall-clock evaluation time — sequential vs. batch-parallel.

14h → 18m

2,000 LIBERO episodes. The bottleneck is environment step rate — the speedup **transfers to any model**.

04 VALIDATION — PITFALLS

One hidden parameter ⇒ catastrophic swings

Each reproduction unmasked an undocumented setting that, alone, moves success rate by tens of points.

- **X-VLA** — wrong proprioceptive state source on LIBERO: **97.8% → 42%** (a 55 pp swing).
- **OpenVLA-OFT** — quaternion antipodal normalization mismatch: LIBERO-Long **95% → 56%**.
- **GR00T** — missing end-effector pose field on SimplerEnv: **30-55% → 0%**.
- **OpenVLA** — undocumented eval-time center crop (scale 0.9): **-3 pp**.

vla-eval records the full configuration with every result — making each run **reproducible from one file**.

05 VLA LEADERBOARD

657 results — one canonical protocol

All Benchmarks (Overview) ▾

Search models...

Published

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Citations ≥ 10

509 models · 17 benchmarks · 657 results · Last updated: 2026-03-04

Paper Coverage by Benchmark

657 results from 509 models · 2685 papers reviewed

Denominator = papers citing the benchmark paper (via [Semantic Scholar](#)). Not all citing papers report evaluation results — this shows how much of that citation pool we have covered.

RLBench	605/791	LIBERO	551/573	CALVIN	391/460	ManiSkill2	260/292	SimplerEnv	268/278
RoboCasa	213/232	RoboTwin 2.0	105/106	VLABench	77/81	RoboTwin 1.0	66/69	Kinetix	52/53
LIBERO-Plus	30/29	RoboArena	23/24	LIBERO-Pro	16/15	MIKASA-Robo	11/10	RoboCerebra	11/10
LIBERO-Mem	4/3	RoboChallenge	2/2						

Model	Params	LIBERO spatial	LIBERO object	LIBERO goal	LIBERO 1D	LIBERO Any-1D	LIBERO 3D	LIBERO-Plus	LIBERO-Pro	LIBERO-Mem	CALVIN	SimplerEnv 0P	SimplerEnv 0P	SimplerEnv wP	ManiSkill2	RLBench
		▾	▾	▾	▾	▾	▾	▾	▾	▾	▾	▾	▾	▾	▾	▾
CosmosPolicy	7B	98.1	100.0	98.2	97.6	98.5	—	—	—	—	—	—	—	—	—	—
Spatial Forcing	—	99.4	99.6	98.8	96.0	98.5	—	—	—	—	—	—	—	—	—	—
EO-1	3B	99.7	99.8	99.2	94.8	98.2	—	—	—	—	—	—	—	—	—	—
RLInf-VLA	7B	98.8	99.8	99.4	94.0	98.1	—	—	—	—	—	—	—	—	—	—
X-VLA	0.9B	98.2	98.6	97.8	97.6	98.1	—	—	—	—	4.430	80.4	75.7	95.8	—	—

Public VLA leaderboard with canonical per-benchmark protocols — filterable by benchmark and by model. Sourced from 1,704 papers that cite at least one tracked benchmark.

657

results across **17 benchmarks** & **509+ models**, AI-extracted from **1,704 papers** with human review of anomalies. Not independently re-run.

Cross-benchmark comparison is rare.

81% of models are evaluated on a single benchmark; only **6%** on three or more.

vla-eval makes cross-benchmark comparison practical — letting the community see **general capability across diverse environments and embodiments**.